THEME Evolution

The Evolutionary History of Maize

Approximately 12,000 years ago, humans began shifting away from a hunter-gatherer way of life and moved toward an agriculturally based one. As early groups of humans began to spend more time in the same location, and sedentary populations grew, they required larger and more readily available food sources. Over time, these early farmers increasingly selected plants that produced greater yields of food with less of an investment in time and energy.

Maize, which we commonly call corn, was first cultivated in Central America nearly 10,000 years ago in the highlands of Mexico. By the time Europeans were exploring Central America in the 1500s, over 300 varieties were being cultivated (Fig. 23B). All of these varieties can trace their ancestral lineage back to a wild Mexican grass known as teosinte.

Among the earliest structural changes that occurred during corn domestication was the loss of *shattering,* the process by which ears of wild corn break apart and disperse their grains (Fig. 23C). A mutation likely caused the ears of some corn plants to remain intact, a trait that is disadvantageous in nature but beneficial to humans. Nonshattering ears are much easier for humans to harvest.



Figure 23B Varieties of corn. Corn can be cultivated to have a specific color, texture, and sugar content. alonsoleon9/iStock/Getty Images

How did our ancestors artificially select for desirable traits? Hunter-gatherers often followed seasonal migration routes between base camps, returning to the same locations year after year. The disturbance of the natural vegetation at these sites would have provided an ideal site for the colonization of species that were being cultivated by these groups. If next season's seeds were collected from the plants that had the most desirable traits, then over time the frequency of plants with these traits would increase. A modern example, showing how quickly artificial selection can change corn, was conducted in 1896 by agricultural scientists selecting for high oil content in corn kernels. Only the top 20% of oil-producing individuals of each generation were allowed to reproduce, and after 90 generations, the average oil content of the corn kernels had increased 450%.

Harvesting plants with specific traits, however, can decrease the genetic diversity within the population. This decline, in turn, produces a genetic bottleneck, in which various traits are lost, while others increase in frequency, giving rise to a change in the phenotypic frequency within the population (see Section 16.1).

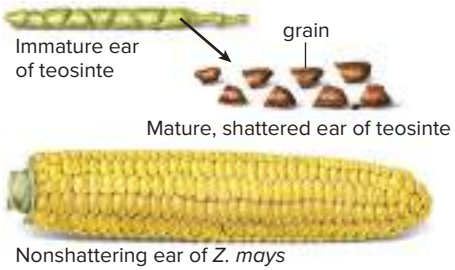


Figure 23C Ears and grain of modern corn and its ancestor, teosinte. Domesticated corn ears are larger than those of the ancestral grass teosinte. In addition, corn fruits are nonshattering, softer, and more edible than are grains of teosinte.

It was the practice of creating corn hybrids in the 1920s that significantly increased yields, as well as the genetic diversity of cultivars. Modern-day corn produces larger grain, generally a more robust plant, an increased amount of apical growth, and fruit with a higher protein and sugar content. In fact, corn has changed so much through artificial selection that domesticated corn can no longer successfully reproduce without human intervention.

Currently, the world's growing population, the loss of farmland, and climate change are straining global food production. The production of corn must now rely on molecular breeding. One method being used to create new strains employs CRISPR/Cas technology that is unique to corn, called the "CRISPR pollen method." This method uses CRISPR-edited genetically modified pollen to carry new versions of a gene to another corn plant's cells. For corn, this method is more effective and versatile than traditional methods of gene modification.

Information gained from genomic studies can also be used to customize corn varieties that increase yields under the growing environmental pressures of drought, pests, and microbial pathogens. Genetic engineering has opened up new windows of evolutionary options for the future of maize and, with that, the future of the humans who rely on this plant.

Questions to Consider

1. What potentially beneficial traits may have been lost during artificial selection in maize?
2. What other crop plants might have undergone a similar form of natural selection?

plants, such as grasses) and most garden plants are flowering plants. This means that all fruits, vegetables, nuts, herbs, and grains, which are the staples of the human diet, are angiosperms. They provide us with clothing, food, medicines, and other commercially valuable products. Over the past 12,000 years, humans have also artificially selected plants to serve us better, resulting in the cultivation of plants as food crops and for other uses. The Evolution feature, "The Evolutionary History of Maize," describes corn (*Zea mays*) as an example.

The flowering plants are called angiosperms because their ovules, unlike those of gymnosperms, are always enclosed within diploid tissues. In the Greek derivation of their name, *angio* ("vessel") refers to the ovary, which develops into a fruit, a unique angiosperm feature.

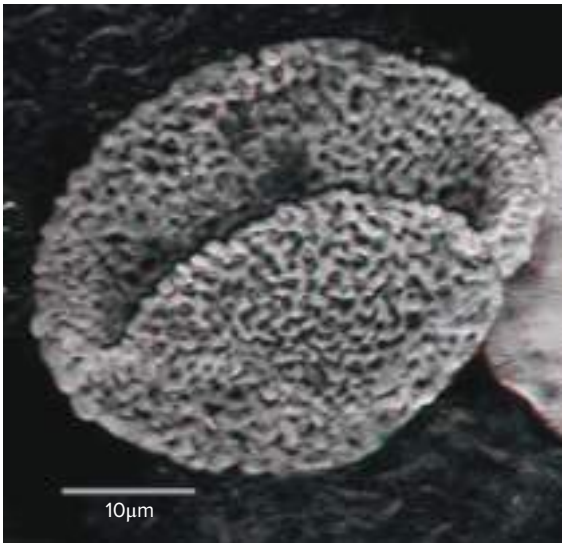
Origin and Radiation of Angiosperms

In general, the fossil record of plants is incomplete, because unlike bones and teeth, the soft parts of plants are often eaten or decayed before they can be preserved. This has created a vigorous debate among paleobotanists on the origins of flowering plants. One of the earliest known angiosperm fossils is

Nanjinganthus dendrostyla, dating back 174 million years to the early Jurassic. *N. dendrostyla* is a stunning fossil with seeds and flowers, giving a glimpse into what some of the early flowering plants may have looked like (see Fig. 19.8). Unlike the soft parts of plants, pollen does preserve well, has characteristic markings, and can be used to date the appearance of a seed plant in an area. Paleobotanists working with drilling cores from Switzerland discovered 240-million-year-old fossilized pollen from the Triassic, pushing the origin of flowering plants back even further (Fig. 23.20*a*).

To find the angiosperm of today that might be most closely related to the first angiosperms, botanists have turned to DNA comparisons. Gene-sequencing data singled out *Amborella trichopoda* (Fig. 23.20*b*) as having ancestral traits. This small, woody shrub, with small, cream-colored flowers, lives only on the island of New Caledonia in the South Pacific. Its flowers are about 4–8 mm wide, and the petals and sepals look the same; therefore, they are called tepals. Plants bear either male or female flowers, with a variable number of stamens or carpels.

Although *A. trichopoda* may not be the original angiosperm species, it is sufficiently close that much may be learned from studying its reproductive biology. Botanists hope that this knowledge will help them understand the early adaptive radiation of angiosperms during the Tertiary period. The gymnosperms were abundant during the Mesozoic era but declined during the mass extinction that occurred at the end of the Cretaceous period (145.5–65.5 MYA). Angiosperms survived and went on to become the dominant plants during modern times.



a. Fossil pollen



b. *Amborella trichopoda*

Figure 23.20 Evolution of flowering plants. **a.** Flowering plant pollen from the Triassic. **b.** Analysis of molecular data suggests that the plant seen here, *Amborella trichopoda,* is the most closely related to the first flowering plants. (a): Susanne Feist-Burkhardt; (b): Doukdouk/Alamy Stock Photo

Monocots and Eudicots

Most flowering plants belong to one of two classes. These classes are the *Monocotyledones,* often shortened to simply the **monocots,** and the *Eudicotyledones,* shortened to **eudicots.** The term *eudicot* (meaning "true dicot") is more specific than the term *dicot.* It was discovered that some of the plants formerly classified as dicots diverged before the evolutionary split that gave rise to the two major classes of angiosperms. These earlier-evolving plants are not included in the designation eudicots.

Cotyledons (Gk. *kotyledon,* "cuplike cavity") are the "seed leaves" containing the nutrients that nourish the plant embryo. If a seed has one cotyledon, it is a monocot, and if a seed has two cotyledons, it is a eudicot (see Fig. 27.8). Several common monocots are corn, tulips, pineapples, and sugarcane; common eudicots include cacti, strawberries, dandelions, poplars, and beans. Table 23.2 lists several fundamental features of monocots and eudicots.

The Flower

Although **flowers** vary widely in appearance (Fig. 23.21), most have certain structures in common (Fig. 23.22).

- The **sepals,** collectively called the calyx, protect the flower bud before it opens. The sepals may drop off or may be colored like the petals. Usually, however, sepals are green and remain attached to the flower stalk.
- The **petals,** collectively called the corolla, are quite diverse in size, shape, and color. The petals are often used to attract a particular pollinator.
- Next are the **stamens.** Each stamen consists of two parts: first, a slender stalk, called a filament, that holds up a second

Table 23.2 Key Features of Monocots and Eudicots

Monocots	Eudicots
One cotyledon	Two cotyledons
Flower parts in threes or multiples of three	Flower parts in fours or fives or multiples of four or five
Pollen grain with one pore	Pollen grain with three pores
Usually herbaceous	Woody or herbaceous
Usually parallel venation	Usually net venation
Scattered bundles in stem	Vascular bundles in a ring
Fibrous root system	Taproot system

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